Dev Agarwal

Software Engineer

+91 7350223740 agarwaldev626@gmail.com linkedin.com/in/devagarwal2503 github.com/devagarwal2503
devagarwal2503.github.io Ahilyanagar, India

PROFESSIONAL SUMMARY

Software Engineer (SDE) with nearly 2 years of experience in building and shipping **production applications**. Strong foundation in **Data Structures** & **Algorithms**, **Object-Oriented Programming**, and **System Design** (C++). Delivered production systems for HRMS, hospitality booking, and e-commerce, including a Leave Management module that reduced manual HR effort by **40%**. Experienced in cross-platform mobile development using **Flutter** and **Dart**, with growing backend expertise in **Node.js**, **Express**, **REST APIs**, and databases like **MongoDB**, **Firebase**, and **Supabase**. Recently developed independent projects featuring AI-integrated tools, GitHub App automation, and a real-time WebSocket system with a custom **Indexed Min-Heap**, actively pursuing SDE roles.

WORK EXPERIENCE

Software Engineer | Trainee — Feb 2026 - Jun 2026
Vishwaguru Infotech — *On-site • Pune, Maharashtra*

- **Designed** and **developed** a comprehensive e-commerce application spanning 8+ screens, including onboarding, authentication, catalog, cart, and payments.
- **Integrated** Firebase Auth, Razorpay, and Supabase for **secure login** and payment processing; achieved promotion to trainee within one month.
- **Created** 10+ responsive UI components and **integrated** 8+ REST APIs, collaborating through OpenProject to ensure timely delivery.

Technologies / Skills Used : *Flutter, Firebase, Supabase, Razorpay, REST APIs, Python, OpenProject*

Software Engineer | Full-time — Mar 2023 - Apr 2024
Centralogic Pvt Ltd — *On-site • Pune, Maharashtra*

- **Delivered** a **Leave Management** module for a production **HRMS** platform with multi-level approvals, reducing manual HR effort by **40%**.
- **Developed** a secure **candidate assessment** platform featuring anti-cheat mechanisms, accommodating **500+** concurrent users.
- **Integrated** over **15 REST APIs** into a hospitality booking application for real-time reservation synchronization.
- **Enhanced** API response handling to minimize data parsing errors and improve application stability.
- **Maintained** code quality through **Git** version control, peer code reviews, and **Agile** sprint delivery using **Jira**.

Technologies / Skills Used : *Flutter, REST APIs, Git, Jira, Agile*

PROJECTS

AI-Powered Code Review Assistant | Github — 2026
Individual Project

- **Developed** a GitHub App that analyzes PRs and posts structured, line-anchored review comments using OpenAI's **gpt-4o-mini**.
- **Identified** context-sensitive issues such as race conditions and cross-file anti-patterns beyond standard linting.
- **Engineered** diff chunking and ±25-line context fetching to maintain large PRs within LLM token limits.
- **Created** a MongoDB-backed React/Vite dashboard that visualizes issues by severity and category, secured via HMAC-verified webhooks.

Technologies / Tools Used : *Node.js, Express, MongoDB, React, Vite, OpenAI API, GitHub Apps & Webhooks, Winston, GitHub Actions*

PulseGrid - Load Balancer Visualizer | Github — 2026
Individual Project

- **Developed** a real-time load balancer simulator to compare **five scheduling algorithms** across a virtual server pool.
- **Created** a custom Indexed Min-Heap with **O(1) peek** and **O(log N) updates**, supporting two of the five scheduling strategies.
- **Engineered** a Predictive Headroom scheduler using **Linear Regression**, reducing CPU load standard deviation from **15.72 to 3.15**, achieving an approximately **80% improvement** compared to WRR.
- **Built** a WebSocket telemetry layer capable of sustaining **10Hz broadcasts** with backpressure-aware frame dropping at a **64KB buffer** threshold.

Technologies / Tools Used : *Node.js, TypeScript, Express, WebSocket (ws), React, Recharts, Vite*

AI Automation Agents — 2026
Individual Project

- Developed an **email summarization automation** processing **50+** incoming emails, generating concise summaries using **LLM APIs**.
- Automated a reply workflow for **sponsorship** and **outreach emails**, reducing manual response time by an estimated **70%** using **n8n** and LLM-based response generation.
- Built **RAG-based AI agents** for **knowledge-retrieval** and conversational automation workflows, with **3** planned implementations.

Technologies / Tools Used : *n8n, LLM APIs, RAG, Workflow Automation*

SKILLS

Programming Languages :	C++, Dart, JavaScript, TypeScript, Python, SQL
Core CS Fundamentals :	Data Structures & Algorithms, Object-Oriented Programming, System Design Basics
Frameworks & Libraries :	Flutter, React, Node.js, Express, Angular
Tools & Platforms :	Git, GitHub, GitHub Actions, Docker, Postman, WebSockets, CI/CD, Linux, Jira, n8n, RAG Agents, LLM API Integration, OpenProject
Databases :	MongoDB, Firebase, Supabase
Soft Skills :	Communication, Leadership, Collaboration, Problem Solving, Time Management
Languages :	English, Hindi, Marathi

EDUCATION

IMS (SPPU) | Ahilyanagar, Maharashrtra — 2022 - 2026
MCA - Master of Computer Applications — *CGPA : 7.38*

Ahmednagar College, SPPU | Ahilyanagar, Maharashrtra — 2019 - 2022
BBA – Computer Applications — *CGPA : 8.81*
Business Administration with Computer Applications - foundation in software engineering and business logic